

Mainly porous Polymers

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Add material that blocks UV and Atomic Oxygen

Add Silica Aerogel

Add Erbium-Doped Silica-Based Glass-Ceramics

Try to make it as cheap and thin as possible.

1. Top/Outer: Ceramic Doped Silica
2. Main shield: Porous Polymer
3. Bottom/Inner: Silica Aerogel

Succeeded test (Test 3) constants:

Ambient Temp

$$T_{amb} = 300 \text{ K}$$

Atmosphere Emmisivity

$$\epsilon_{atn} = 0.75$$

Angle

$$\cos(\theta) = 1$$

Thermal emissivity

$$\epsilon = 0.96$$

Solar Absorptivity

$$\alpha = 0.015$$

Conductivity

Silica/Ceramics

$$k_{ceramic} = 1.4 \text{ W/m} \cdot \text{K}$$

Erbium-Doped Glass Equivalent

$$k_{g;ass} = 1.2 \text{ W/m} \cdot \text{K}$$

Porous Polymer

$$k_{polymer} = 0.03 \text{ W/m} \cdot \text{K}$$

Aerogel

$$k_{aerogel} = 0.02 \text{ W/m} \cdot \text{K}$$

Thickness

$$L_1 = 4 \times 10^{-6} \text{ m}$$

$$L_2 = 1.2 \times 10^{-5} \text{ m}$$

$$L_3 = 1 \times 10^{-4} \text{ m}$$

$$L_4 = 8 \times 10^{-5} \text{ m}$$

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Refractive Index

$$n \approx 1.45$$

Regular particle size

$$r \approx 0.2 \text{ Micrometer}$$

Test 3 (Succeeded)

Absorbed Solar Irradiance Equation

$$Q_{abs} = \alpha \cdot G_{sc} \cdot \cos(\theta)$$

$$Q_{abs} = 0.015 \cdot 1361$$

$$Q_{abs} = 20.24 \text{ W/m}^2$$

Stefan Boltzmann Law Atmosphere

$$P_{atm}(T) = \epsilon_{atm} \cdot \sigma \cdot T_{amb}^4$$

$$P_{atm}(T) = 0.75 \cdot 5.67 \times 10^{-8} \cdot (300^4)$$

$$P_{atm} = 344 \text{ W/m}^2$$

Stefan Boltzmann Law

$$P_{rad}(T) = 0.96 \cdot 5.67 \times 10^{-8} \cdot T^4$$

Radiative Heat Balance Equation

$$P_{net}(T) = P_{rad}(T) - P_{atm} - Q_{abs}$$

$$P_{net}(T) = 0.96 \cdot 5.67 \times 10^{-8} \cdot T^4 - 344 - 20.42$$

$$P_{net}(T) = 0.96 \cdot 5.67 \times 10^{-8} \cdot T^4 - 364.42$$

Radiative Thermal Equilibrium

$$P_{net}(T) = 0$$

$$0.96 \cdot 5.67 \times 10^{-8} \cdot T^4 = 364.42$$

$$T^4 = \frac{364.42}{0.96 \cdot 5.67 \times 10^{-8}}$$

$$T^4 = 6.7 \times 10^9$$

$$T = 285.9 \text{ K}$$

Cooling Check

$$P_{rad}(300) = 0.96 \cdot 5.67 \times 10^{-8} \cdot (300^4)$$

$$P_{rad}(300) = 441 \text{ W/m}^2$$

$$P_{net} = 441 - 344 - 20.42$$

$$P_{net} = 76.6 \text{ W/m}^2$$

$$P_{net} > 0 \rightarrow \text{Net Radiative Cooling}$$

$$\Delta T = 300 - 285.9 = 14.1$$

Wien's Law

$$\lambda_{peak} = \frac{2.897 \times 10^{-3}}{285.9}$$

$$\lambda_{peak} = 10.14 \text{ Micrometer}$$

Thermal Resistance Equation

Aerogel layer

$$L = 8 \times 10^{-5}, k = 0.02 \rightarrow R = 4 \times 10^{-3}$$

Foam layer

$$L = 1 \times 10^{-4}, k = 0.03 \rightarrow R = 3.33 \times 10^{-3}$$

Series Resistance Equation

$$R_{total} = 4 \times 10^{-3} + 3.33 \times 10^{-3}$$

$$R_{total} = 7.33 \times 10^{-3}$$

Fourier's Law

$$\Delta T = 300 - 285.9$$

$$\Delta T = 14.1$$

$$q = \frac{14.1}{7.33 \times 10^{-3}}$$

$$q = 1923 \text{ W/m}^2$$

Radiative Cooling Requirement

$$\epsilon \sigma T^4 > Q_{abs} + P_{atm}$$

$$440 > 364.42$$

Succeeded.